

SkillNest — Phase 2 Technical Checklist

What is required after Phase 1 · Target: unique, local (India-first), impressive beyond Fiverr/Upwork

A. Infra & DevOps (foundation)

ITEM	WHY (BUSINESS REASON)	HOW	EFFORT
1. Production queue (DB ? Redis) + workers	Emails, webhooks, AI jobs, notifications block the request today (queue=sync). Needed for reliable order mail, PDFs, scoring.	Enable Redis (Laravel 9 native), set <code>QUEUE_CONNECTION=redis</code> , run <code>php artisan queue:work</code> under Supervisor; optional Horizon (L9 supports Horizon for Redis).	M
2. Scheduler (cron)	Auto escrow release timers, seller availability day rollover, KYC reminder, wallet settlement, invoice reminders.	<code>* * * * * php artisan schedule:run</code> ; write task classes in <code>app/Console</code> .	S
3. Webhook relay (outgoing)	Push order/payment events to Telegram/Slack/Discord for seller alerts; post to client's system.	Custom <code>WebhookDeliveryService</code> with retry + signature; or Laravel 9 outbound package; DB table <code>webhook_deliveries</code> .	M
4. Webhook ingress (incoming)	Accept third-party callbacks: payment IPN (CCAvenue), eKYC/Digilocker, SMS DLR, WhatsApp status.	Signed routes + <code>VerifyCsrfToken</code> whitelist; HMAC/secret validation; idempotent processing jobs.	M
5. Object storage / CDN	Uploads (KYC docs, order docs, media) scale beyond one disk; public CDN for images.	MinIO self-host or S3-compatible (q 2026: AWS/GCS/B2); keep local disk fallback.	M
6. Search engine	Replace slow LIKE search on gigs for typo-tolerant, fast results (jobs, services, sellers).	Meilisearch (easy; PHP client) or OpenSearch. Sync via jobs on service save.	M
7. AI Gateway	Single place to call LLM providers (OpenAI / Claude / Gemini / local) for suggestions, matching, moderation, translation.	Laravel `Ai` microservice OR curated package; config API keys; rate limits + cost guardrails.	S-M
8. Monitoring & alerting	Know when payments/webhooks break; error tracking in prod.	Sentry or Flare; Laravel PULSE for queue/cache (L9: use <code>spatie/laravel-health</code> or <code>pulse</code> alternative); UptimeRobot.	S
9. Cache + session to Redis	Multi-server ready, faster settings/catalog reads (currently <code>cache=file</code> , <code>session=file</code>).	<code>REDIS_HOST</code> /port config; <code>SESSION_DRIVER=redis</code> , <code>CACHE_STORE=redis</code> .	S
10. CI/CD pipeline	Release webhooks/microservices/AI safely instead of manual deploy.	GitHub Actions (lint, migrate check, build, tests) + deploy via <code>rsync/Enovadeployer</code> or Docker.	M

11. Logging pipeline	Correlate requests across monolith + microservices (trace id).	JSON logs ? single file or Loki/S3; add request_id middleware.	S
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B. Microservices (when do they split out)

SERVICE	RESPONSIBILITY	STACK	COMMS
notification-service	Email + SMS + WhatsApp + push, single deliverable API (de-dup, receipt); takes load off main app.	Laravel or Node; MSG91/Fast2SMS, WhatsApp Business API, Expo/OneSignal push.	HTTP + Redis queue
ai-service	Translation (EN/HI), recommendation/matching, quote suggestion, moderation, chat assistant.	Python FastAPI; FastText/Whisper for local, LLM APIs for heavy.	HTTPS JSON + async jobs
search-service (optional)	Index & query gigs/sellers; typo-tolerance, geo filters.	Meilisearch/OpenSearch behind small API.	HTTP
media-service (optional)	Image/PDF processing at scale (thumbnails, watermarks, KYC doc optimization).	ImageMagick/ffmpeg workers or imagedelivery (Cloudflare).	Queue
payment-service (optional)	Extract current PaymentService + drivers into independent deployable; multi-currency later.	Laravel, kept same shape (PaymentGatewayManager).	HTTP/webhooks

C. AI Capabilities (features + what's needed)

AI FEATURE	WHY UNIQUE/VALUABLE	REQUIREMENT	EFFORT
Service suggestions	Already planned in P1 (api.ai.suggest-services) — enhance with buyer profile + history.	ai-service + redis user profile cache.	M
Auto-quote assistant	Buyer pastes requirement ? seller gets fair price + scope + delivery estimate (huge for non-tech local services).	LLM prompt + service catalog context + price history of similar orders.	M
Local language (Hinglish/Hindi)	India-first; massive gap vs Fiverr/Upwork. Translate listings, requirements, chat.	LLM translation call or IndicTrans lib; store t on demand, cache.	M
Voice-to-requirement	Non-technical users (maids, plumbers) can speak the job; app converts to text/structured job.	Whisper (local) or cloud STT; attach to order requirement builder.	M
Smart matching / recommendation	Buyer sees best sellers for job; boosts conversion.	RAG over services+sellers, click/conv signals, rankings.	L
AI moderation & fraud triage	Detect scammy gigs, chats, reviews before humans; auto-dispute "first look".	Classifiers on listings/chat/reviews; escalate to admin.	M

Pricing intelligence	Seller gets suggested price from category/market data (gold for non-tech sellers).	Aggregate completed orders + analytics service.	M
Super-customer support copilot	Answer FAQs / disputes instantly (India support cost killer).	Support KB + LLM (chat UI + agent inbox).	M

D. Webhook / Integration Events (Phase 2 wiring)

EVENT	TRIGGER	DESTINATION / USE	TYPE
order.created	Buyer places order (incl. `force` while seller offline)	Telegram/Discord seller alert, client CRM (outgoing).	out
order.paid / payment.captured	Gateway callback (CCAvenue/Demo) marks paid	Seller notify + webhook to client; escrow release start.	in+out
order.accepted / delivered / cancelled	Seller/buyer transitions	SMS/WhatsApp buyer + webhook.	out
project.milestone	Milestone delivered	Buyer timeline push + webhook.	out
kyc.status	Admin verifies/rejects	Notify seller; re-upload flow.	in
sms.dlr / whatsapp.delivery	Provider status callbacks	Re-send/retry logic in notification-service.	in
ekyc.webhook	Digilocker / UIDAI consent result	Auto-fill + verify seller KYC (Aadhaar eKYC) — big trust differentiator.	in

E. Third-party Integrations

INTEGRATION	PURPOSE	PROVIDER OPTIONS	EFFORT
SMS / transactional	Order + OTP alerts (email alone misses local sellers).	MSG91, Fast2SMS, Twilio	S
WhatsApp Business API	Seller/buyer chat + order notifications — India primary channel.	Meta WABA via Gupshup/360dialog/GreenAPI	M
Payments (expand)	UPI auto-collect, cards, netbanking, EMI; settlement.	CCAvenue (P1), Razorpay, Cashfree, Stripe for intl.	M
eKYC (Aadhaar/Digilocker)	Digital seller verification w/ consent — huge trust + conversion.	DigiLocker API, UIDAI eKYC (licensed partner)	M
E-tax / invoicing	GST invoices, TDS handling for services.	IRP (e-invoice), Zoho Books custom	M
AI providers	LLM for AI features.	OpenAI / Anthropic / Gemini / local Ollama on GPU box	S

Generated docs/phase-2-listing.html · Companion docs: phase-2-roadmap.html (big-picture plan) & server-configuration-roadmap.html (<what + why>). E=Eye estimates: S<1 wk, M 1–3 wk, L 4–8 wk per feature, sequential team-of-one.